OFFICIAL PLAYER MANUAL · V1.0 · 2026

VR NEXUS

SILICON & SENSES

A high-fidelity corporate strategy simulation. Build your startup, crush the duopoly, and dominate the VR market across 8 financial quarters.

Finance for Non-Financial Decision Makers · Simulation Module

CHAPTER 01

Game Overview & Objective

It is **2032**. The VR market is dominated by two entrenched giants: **Apex Optic**, the elite gatekeeper of premium immersion, and **ValueVirtua**, the manufacturer of budget, low-quality headsets. The world is hungry for a third option.

You are the founder of a stealth-mode startup that has just secured a **$500,000 seed round**. Over 8 financial quarters, your team must design a disruptive VR headset, manage volatile cash flow, and outmaneuver both AI competitors and rival player-companies to claim market dominance.

🏆 The Winning Condition

The company with the highest **Weighted Score** at the end of Quarter 8 wins. The score rewards long-term financial health — not short-term revenue spikes.

$$\text{Score} = (0.4 \times \text{Total Equity}) + (0.4 \times \text{Market Share}) + (0.2 \times \text{Brand Equity})$$

40% — TOTAL EQUITY

Shareholders' Equity + Retained Earnings

40% — MARKET SHARE

Units Sold / Total Market Units

20% — BRAND EQUITY

Accumulated Brand Reputation Points

⏱️ Starting Conditions

Parameter	Starting Value
Cash	$500,000
Fixed Assets (Gross)	$200,000

Parameter	Starting Value
Component Level	Level 1 (all four components)
Inventory	0 units
Brand Equity	10 pts
Debt	$0
Max Production Capacity	2,000 units / quarter

🔄 The Game Loop (Every Quarter)

1. **Analyze** — Review your P&L, Cash Flow Statement, Balance Sheet, and Market Intelligence.
2. **Decide** — Set your Price, Production Volume, Brand Spend, CapEx investment, R&D upgrades, and financing moves.
3. **Execute** — Lock in decisions. The GM triggers the engine, which calculates sales, market share, and updates the ledger.
4. **Debrief** — The GM leads a "War Room" session reviewing results, efficiency metrics, and competitive benchmarks.

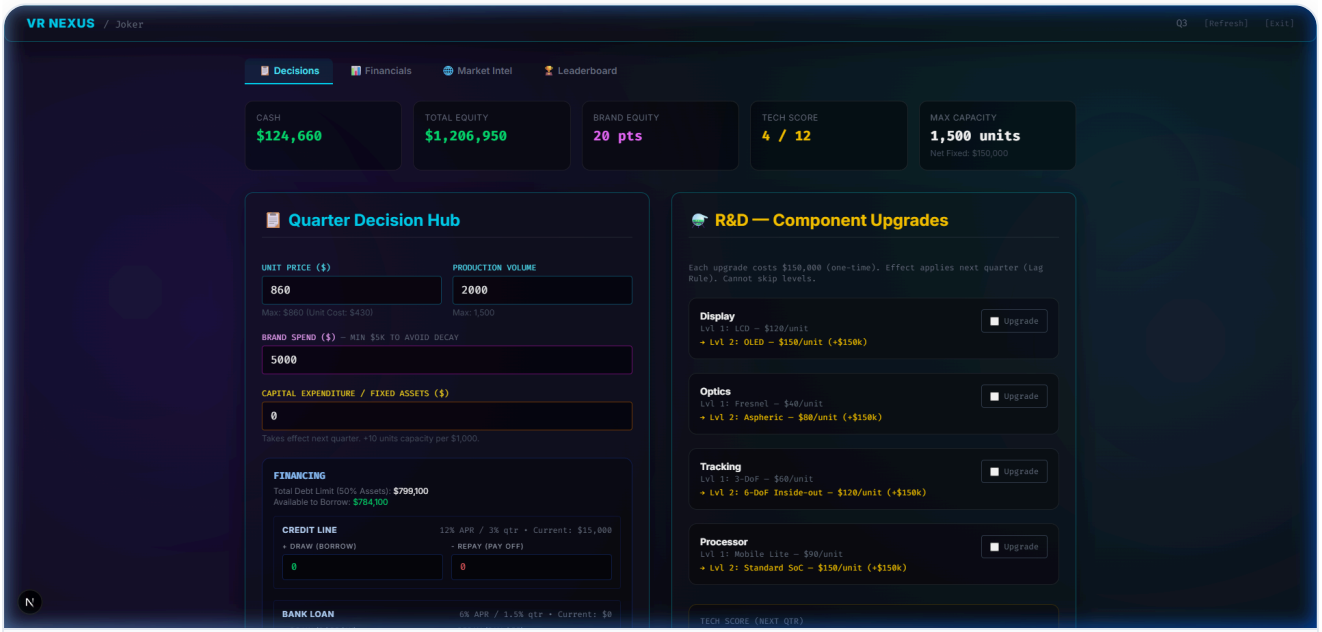
CHAPTER 02

The Dashboard — Your Command Center

Accessing the simulation begins at the **Welcome Screen**. Enter your **Company ID** (provided by your Game Master) to access your private dashboard. The Game Master accesses a separate control panel via a secret password.

📋 Decisions Tab — The Quarter Decision Hub

This is where every strategic decision is made each quarter. The dashboard is split into two panels: the Decision Hub (left) and the R&D Upgrade Panel (right).



DECISIONS TAB — Quarter Decision Hub with KPI strip, pricing controls, brand spend, CapEx, financing, and R&D upgrade panel

🔹 KPI Status Strip (top bar)

Five key metrics update in real-time as you configure your decisions:

Metric	Color	What It Tells You
Cash	Green	Current liquid capital. Never let this reach $0.
Total Equity	Cyan / Red	Your core winning metric. Green if above seed capital.

Metric	Color	What It Tells You
Brand Equity	Fuchsia	Your accumulated reputation, drives the Attractiveness Score.
Tech Score	Yellow	Sum of all component levels. Ranges from 4–12.
Max Capacity	White	The ceiling on units you can build this quarter.

Pricing & Production

Set the **Unit Price ($)** for your headset and the **Production Volume** (capped by your Max Capacity). The interface will warn you in real-time if your price exceeds the allowed maximum for your current Tech Score tier.

> **PRICING RULE**
> Your maximum allowable price is calculated as: **Unit Build Cost × Tier Multiplier**. Budget (Score 4–5): ×2.0, Mid-Range (6–8): ×2.5, Premium (9–12): ×3.0.

Brand Spend

Allocate your quarterly investment in brand marketing. **Every $1,000 spent earns 1 Brand Equity point.** Spending less than $5,000 in any quarter triggers a 5% brand decay on your existing equity balance.

Capital Expenditure (CapEx)

Invest in factories and assembly lines to grow your production ceiling. Every **$1,000 of CapEx adds 10 units of quarterly capacity** — but this only takes effect in the *following* quarter (the Lag Rule).

Financing Panel

Two financing instruments are available. Use them thoughtfully — excessive debt triggers solvency penalties.

Instrument	Rate	Best Used For
Revolving Credit Line	12% APR / 3% per qtr	Short-term cash flow gaps (inventory timing)
Bank Loan	6% APR / 1.5% per qtr	Long-term capital investment (CapEx, factories)

> **THE 50% RULE — LIQUIDITY FREEZE**
> Total Debt cannot exceed **50% of Total Assets**. If it does — or cash hits $0 — you enter a **Liquidity Freeze**: all Marketing and R&D spending is blocked until solvency is restored. You may

> still produce and sell, but you're fighting with one hand tied behind your back.

🔹 R&D — Component Upgrades Panel

On the right side of the Decisions tab, you can initiate upgrades to any of your four headset components. Each upgrade costs a one-time **$150,000 R&D fee**. Upgrades cannot skip levels and take effect in the next quarter.

CHAPTER 03

Product & Components

Every headset is an assembly of four core components. Your choices determine your **Tech Score**, **Unit Build Cost**, and ultimately your **market attractiveness**.

Component	Level 1 – Budget	Level 2 – Mid-Range	Level 3 – Premium
Display	LCD — **$120/unit**	OLED — **$150/unit**	Micro-OLED — **$250/unit**
Optics	Fresnel — **$40/unit**	Aspheric — **$80/unit**	Pancake — **$130/unit**
Tracking	3-DoF — **$60/unit**	6-DoF Inside-out — **$120/unit**	6-DoF + Eye Tracking — **$180/unit**
Processor	Mobile Lite — **$90/unit**	Standard SoC — **$150/unit**	High-Performance — **$220/unit**

Plus a flat **$120/unit Assembly Cost** added to all configurations.

Tech Score

Each component contributes Tech Points equal to its level number (Level 1 = 1pt, Level 2 = 2pts, Level 3 = 3pts). Your Tech Score is the sum.

```
Tech Score = Display Lvl + Optics Lvl + Tracking Lvl + Processor Lvl  (Range: 4 - 12)
```

The Lag Rule — Nothing Is Instant

All decisions take effect the quarter *after* they are submitted. Plan ahead!

Decision Made in Q*n*	Takes Effect
R&D Upgrade	Start of Q*n+1* — new component available for production
Production Volume	Units built in Q*n* are available to sell in Q*n+1*
CapEx (Fixed Asset purchase)	New capacity operational from Q*n+1*

Decision Made in Q*n*	Takes Effect
Loans / Credit Draws	Cash available from Q*n+1*

R&D Maintenance Cost

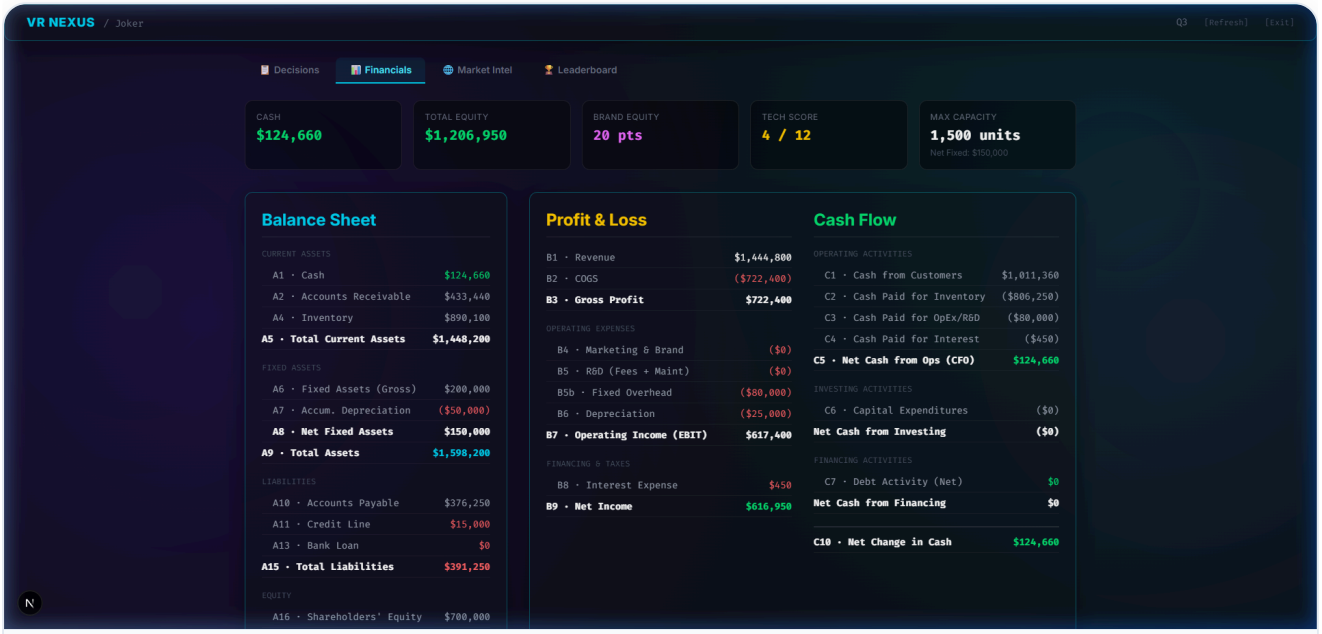
Advanced components require ongoing engineering effort. For every component level above Level 1, you pay a **recurring $15,000/quarter maintenance fee**. This is a direct charge on your P&L every quarter.

> 💡 **EXAMPLE**
>
> Two components at Level 2, one at Level 3 = (2×$15k) + (1×$15k) = **$45,000/quarter** in R&D Maintenance, before any new upgrades.

CHAPTER 04

Financial Statements

The simulation generates three professional-grade financial reports after every quarter. Mastering how to read them is central to winning.



FINANCIALS TAB — Live Balance Sheet (left), Profit & Loss and Cash Flow Statement (center / right), and Historical Trend Charts (below)

A. Balance Sheet

A snapshot of your company's financial position at the end of any quarter.

Line	Item	Notes
A1	**Cash**	Liquid capital. The lifeblood of your operations.
A2	Accounts Receivable (AR)	30% of previous quarter's sales revenue, not yet collected.
A4	Inventory	Unsold units valued at Unit Build Cost (FIFO).
A5	**Total Current Assets**	A1 + A2 + A4
A8	**Net Fixed Assets**	Gross Fixed Assets – Accumulated Depreciation. **Drives production capacity.**

Line	Item	Notes
A9	**Total Assets**	A5 + A8
A10	Accounts Payable (AP)	50% of current quarter's component costs, deferred.
A11	Credit Line	Flexible high-interest revolving debt.
A13	Bank Loan	Fixed-term low-interest debt.
A18	**Total Equity**	Shareholders' Equity + Retained Earnings. **40% of your winning score.**

✅ **BALANCE SHEET CHECK**

Total Assets (A9) must always equal Total Liabilities + Total Equity (A15 + A18). Any discrepancy indicates a calculation error.

B. Profit & Loss (P&L) Statement

Line	Item	Calculation
B1	**Revenue**	Units Sold × Unit Price
B2	COGS	Units Sold × Unit Build Cost
B3	**Gross Profit**	B1 − B2
B4	Marketing & Brand Spend	OpEx — direct cash out
B5	R&D Expenses	One-time Upgrade Fees + Recurring Maintenance ($15k × levels above 1)
B5b	Fixed Overhead	**$80,000/quarter** — non-negotiable factory & ops cost
B6	Depreciation	**12.5% of Gross Fixed Assets per quarter** (capped at 80% of asset value)
B7	**Operating Income (EBIT)**	B3 − (B4+B5+B5b+B6)
B8	Interest Expense	(Credit Line × 3%) + (Bank Loan × 1.5%)
B9	**Net Income**	B7 − B8

C. Cash Flow Statement

Profit is a theory; cash is a fact. The cash flow statement reconciles these two, accounting for the timing gap between accrual accounting and actual cash movement.

Line	Item	Formula
C1	Cash from Customers	(Current Revenue × 70%) + Prior Quarter AR
C2	Cash Paid for Inventory	(Current Build Costs × 50%) + Prior Quarter AP
C5	**Net Cash from Ops (CFO)**	C1 − (C2 + C3 + C4)
C6	Capital Expenditures	Cash spent on Fixed Assets this quarter
C9	Cash Flow from Financing	Debt draws − Principal repayments
C10	**Net Change in Cash**	C5 − C6 + C9

CHAPTER 05

The Market Engine & Competitors

How Demand is Calculated

The simulation uses a **Logit Demand Model**. Every company's sales are governed by an **Attractiveness Score (A)**, which captures the price-to-performance ratio amplified by brand reputation:

$$A = (\text{Tech Score} / \text{Price}) \times \text{Brand Equity}$$

Your share of the total market is then proportional to your attractiveness vs. all competitors:

$$\text{Company Market Share} = A_{\text{yours}} / \Sigma A_{\text{all companies}}$$

Units Sold = Company Market Share × Total Market Size, capped by your available inventory. If you run out of inventory (a "stockout"), unmet demand spills over to your next most attractive competitor.

The total market starts at **10,000 units/quarter** and grows at a rate set by the GM (default: **7% per quarter**).

Your AI Competitors

Apex Optic PREMIUM AI

- **Tech Score:** 12 (all Level 3)
- **Starting Price:** $1,800/unit
- **Brand Equity:** 150 pts
- **Market Share:** ~35% initially
- Never drops price below $1,500
- Boosts brand spend if you reach Tech Score 11+

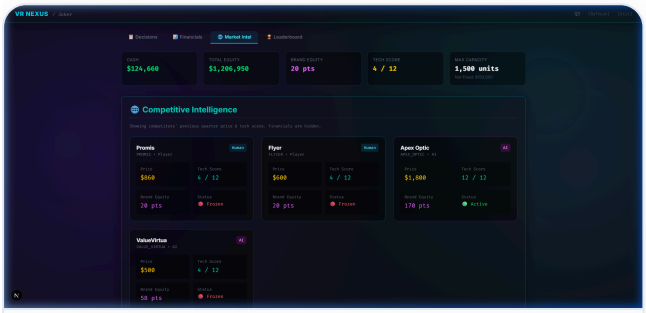
ValueVirtua BUDGET AI

- **Tech Score:** 4 (all Level 1)
- **Starting Price:** $500/unit
- **Brand Equity:** 60 pts
- **Market Share:** ~30% initially
- Matches your price if within 10% of its own
- Cuts price by $10/qtr if share falls below 15%

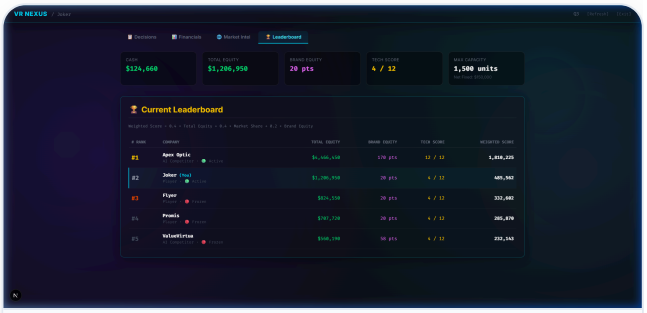
📌 **OPENING MARKET**

The remaining ~35% of the 10,000-unit market is split among all player companies in Q1 based on their opening Attractiveness Scores. This is your beachhead.

Market Intelligence & Leaderboard



MARKET INTEL — Competitor price, Tech Score, Brand Equity & market conditions

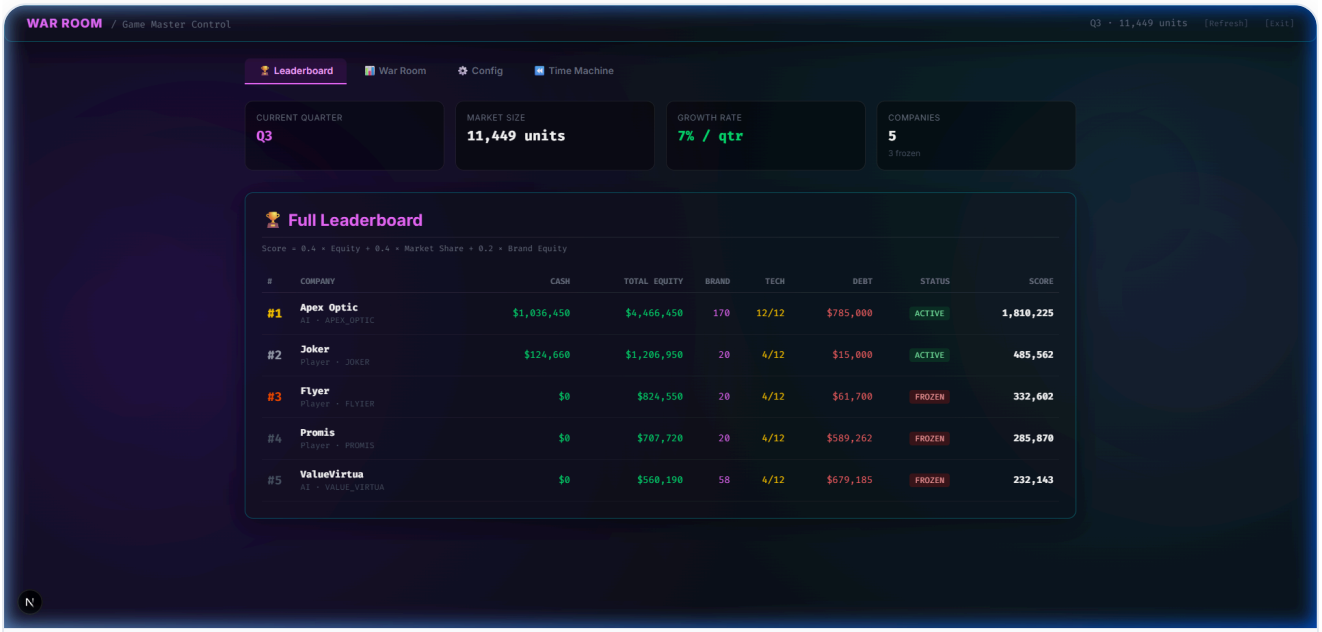


LEADERBOARD — Live rankings by Weighted Score, Total Equity & Brand Equity

CHAPTER 06

The Game Master & War Room

Between quarters, the Game Master leads a structured **War Room Briefing**. This is the core educational session of the simulation — a structured debrief that connects every team's financial decisions to their market outcomes.



GM WAR ROOM — Leaderboard, configuration controls, market shock triggers, and rollback capability

War Room Agenda

1. **The Leaderboard** — Rankings by Total Equity, Market Share, and Weighted Score are revealed.
2. **Competitive Benchmarking** — The GM compares Fixed Asset Turnover, Inventory Days, and Gross Margin side-by-side.
3. **Market Shocks** — Unexpected external events are announced (e.g., chipset shortages, OLED price drops, competitor brand scandals).
4. **Pivot Discussion** — The GM spotlights a Cash Flow success story or a Debt Spiral cautionary tale.
5. **Shadow KPI Review** — If a rollback occurred, the GM compares the "Old Timeline" vs. "New Timeline" outcomes for a structured "What If" discussion.

External Shock Events (Examples)

Quarter	Shock Event	Impact
Q3	Global chipset shortage	Level 3 Processor cost +20% for one quarter
Q5	OLED yield breakthrough	Level 2 Display cost −15% permanently
Q6	Consumer advocacy report	ValueVirtua's Brand Equity halved

CHAPTER 07

Strategies & Pro Tips

The following tips are distilled from the rules of the simulation engine itself. Understanding the "why" behind each recommendation is the key to truly winning.

Plan One Quarter Ahead — Always

The Lag Rule is the most important mechanic in the game. CapEx won't expand your capacity this quarter; a drawn loan won't give you cash right now. Every strategic move must be made one quarter before you actually need it. Reactive decision-making is a losing strategy.

Never Let Brand Equity Decay

Brand Equity is a multiplier in the Attractiveness Formula. A great Tech Score means nothing if your brand reputation is eroding. Maintain at least $5,000/quarter in Brand Spend at all times — even when cash is tight. Sacrificing brand for one quarter can cost you multiple quarters of market share.

Expand Cautiously — Depreciation Is Brutal

Every $100,000 of Fixed Assets generates $12,500 of depreciation expense per quarter, regardless of sales. Aggressive factory expansion without matching revenue volume will crater your P&L. Scale production capacity in line with realistically achievable demand.

Match Debt Maturity to Asset Maturity

Use the
Revolving Credit Line (12% APR)
for short-term operational gaps only. Use
Bank Loans (6% APR)
for long-term CapEx investment. Financing a factory purchase with a revolving credit line will cause runaway interest costs that are nearly impossible to recover from.

Avoid Both Stockouts and Pile-ups

Excess inventory is cash tied up on a shelf — it appears on your Balance Sheet but is not earning revenue. A stockout means you lost sales to your competitors. Study the Market Intel tab before committing production volume. Produce for demand, not factory capacity.

Find the Gap Between the AI Titans

Apex Optic (Tech Score 12, $1,800) and ValueVirtua (Tech Score 4, $500) deliberately leave a vacuum in the mid-range ($700–$1,200) and mid-tech (Score 6–9) tier. A Med-Range headset with strong brand investment will out-attract both AIs for a large portion of price-sensitive but quality-conscious consumers.

Watch Cash Flow — Not Just Net Income

A company can be profitable on the P&L while running out of cash due to inventory build-up and receivables timing (the AR/AP gap). Always check the Cash Flow Statement (C10 Net Change in Cash) before celebrating a profitable quarter.

Total Equity Is the Real Scoreboard

The winning formula weights Total Equity at 40%. Strategies that generate consistent Net Income (which flows into Retained Earnings) compound over 8 quarters. A high-revenue but high-cost strategy that generates no retained earnings will always lose to a lean, efficient competitor.

APPENDIX

Key Terms Glossary

Attractiveness Score (A)	The primary demand driver: (Tech Score / Price) × Brand Equity. Determines your relative market share via the Logit Model.
Brand Equity	Accumulated reputation points. Gains 1 pt per $1K of brand spend; decays 5% if spend is below $5K. Capped at 200 pts.
CapEx	Capital Expenditure. Cash invested in Fixed Assets (factories). Increases production capacity by 10 units per $1K spent (next quarter).
Depreciation	Non-cash expense of 12.5% of Gross Fixed Assets per quarter. Reduces Net Income but not Cash. Capped when A. Depreciation = 80% of asset value.
EBIT	Earnings Before Interest and Taxes. Also called Operating Income (B7). Revenue minus all operating costs, before financing costs.
Fixed Overhead	$80,000 per quarter non-variable cost covering factory rent, utilities, and non-R&D staff. Cannot be reduced.
The Lag Rule	All decisions submitted in Quarter N take effect at the start of Quarter N+1. Nothing is instantaneous.
Liquidity Freeze	Triggered when Total Debt > 50% of Total Assets, or Cash = $0. Marketing and R&D spending are blocked until solvency is restored.
Logit Demand Model	The mathematical model that converts Attractiveness Scores into market share percentages. Your share = your A / sum of all companies' A.
Net Fixed Assets (A8)	Gross Fixed Assets – Accumulated Depreciation. Directly drives Max Production Capacity (Net Fixed / 100 = max units/quarter).
R&D Maintenance Cost	$15,000/quarter per component level above Level 1. Recurring P&L cost for operating advanced technology.
Retained Earnings (A17)	Cumulative Net Income from all previous quarters. A key component of Total Equity and thus the Weighted Winning Score.

Stockout	When your demand exceeds your inventory. Unmet customer demand spills over to your most-attractive competitor.
Tech Score	Sum of all four component levels (range: 4–12). Determines your product tier and price multiplier ceiling.
Total Equity (A18)	Shareholders' Equity + Retained Earnings. The single most important metric — 40% of your final winning score.
Weighted Score	The final tiebreaker: 0.4 × Equity + 0.4 × Market Share + 0.2 × Brand Equity. Calculated at end of Q8.

This manual is confidential and intended for registered simulation participants only.